UNIVERSITY OF LIVERPOOL LIV.INNO

Alma 9 Validation: Single Muon MC Overview

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FASER CERN

Overview

- Aim:
  - Validate new tracking in migration of offline software to Alma 9
- Samples looked at as part of this study are MC
  - Both samples are particle gun - single muon samples [+ve and -ve muons] produced for the FLUKA energy and angular spectra
    - 100117 (/eos/experiment/faser/data0/sim/mc24/particle_gun/100117/phy/
    - 100116 (/eos/experiment/faser/data0/sim/mc24/particle_gun/100116/phy/
  - New tracking samples (s0012-dev) are compared to old tracking samples (s0012-r0019) that have the same MC sample identifier
- As part of the validation looking at :
  - Getting resolutions using residuals
  - Track efficiency
  - Charge misidentification etc.

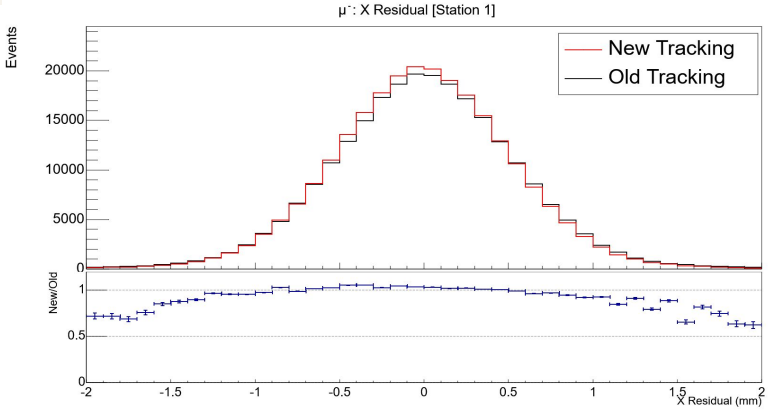
Data Selection

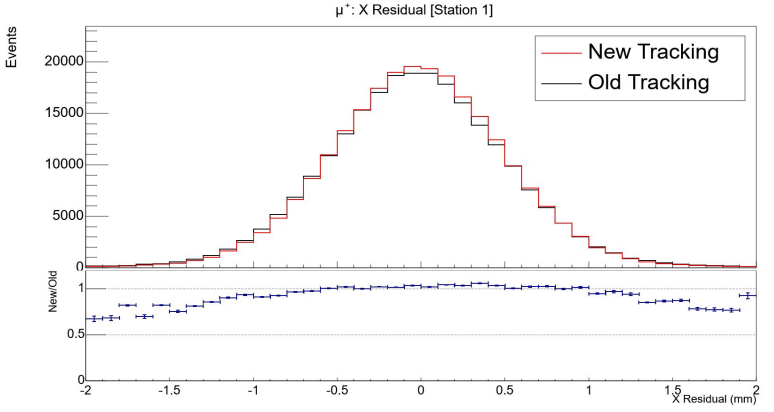
- Applied the following cuts to the data:
  - Fiducial truth in all 3 stations (& at vertex)
  - r_atMaxRadius < 100
  - Exactly one 1 Good Track - *See twiki for definition of a “good” track*
  - Matching using:
    - PDG value [for truth]
    - Track charge [for reco]

Residuals

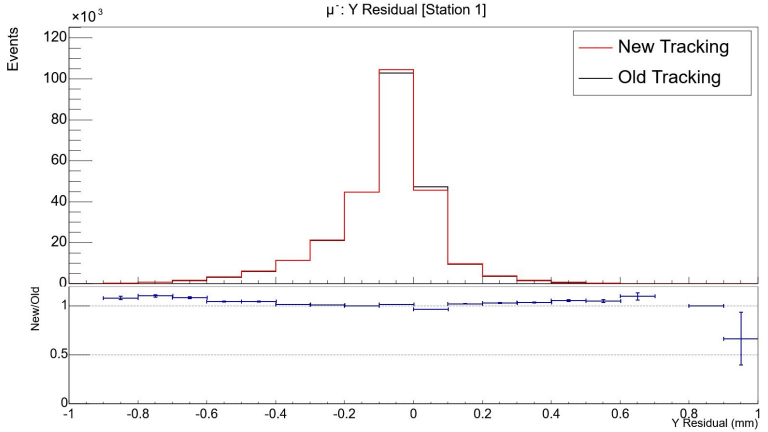
Residuals - X Positions

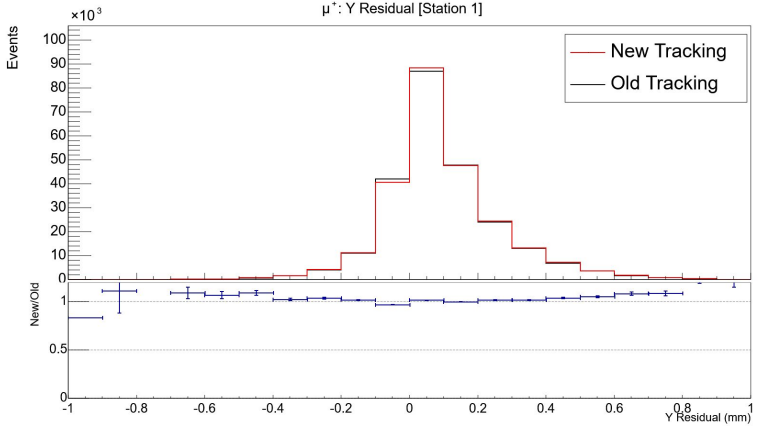
- Residual defined as difference between truth and reco
  - E.g. for residual of X position : $XPosition_{Truth@St1}$-$XPosition_{Reco@St1}$
  - Note: For truth positions - these are an averaged truth position over the whole station - unlike the reconstructed track
  -



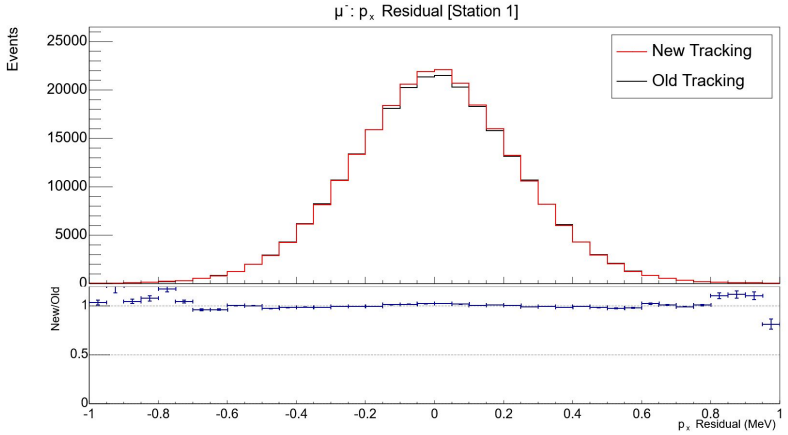


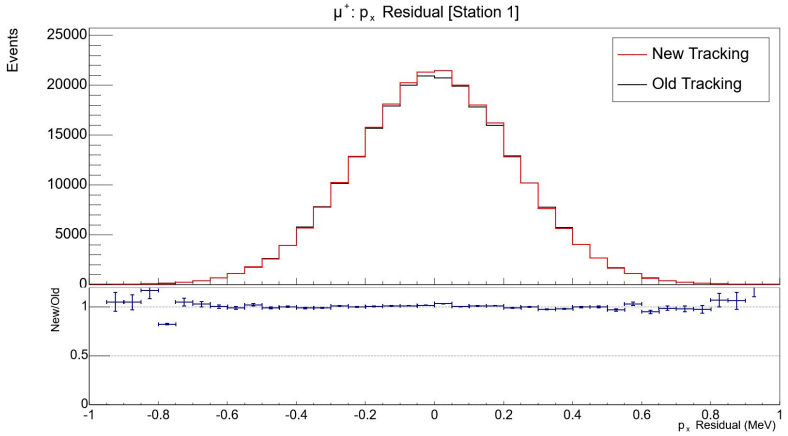
Residuals: Y Positions



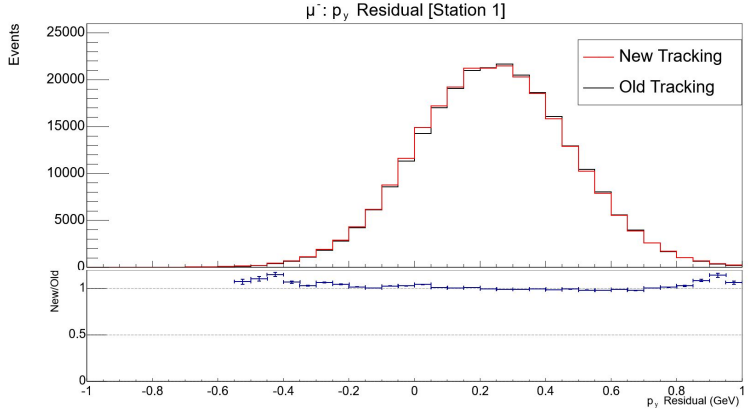


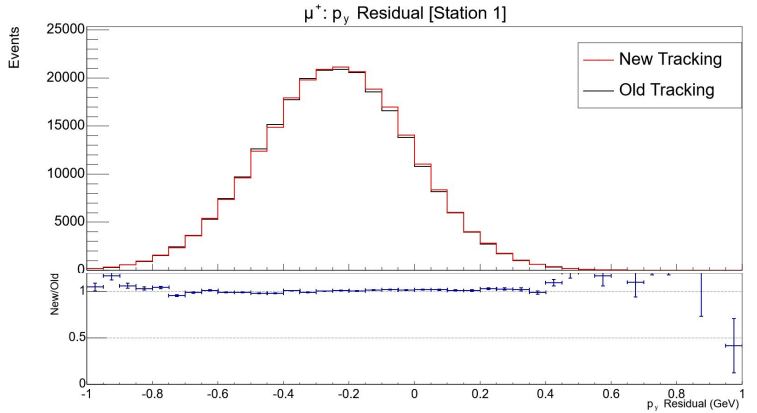
Residuals: X Momentum



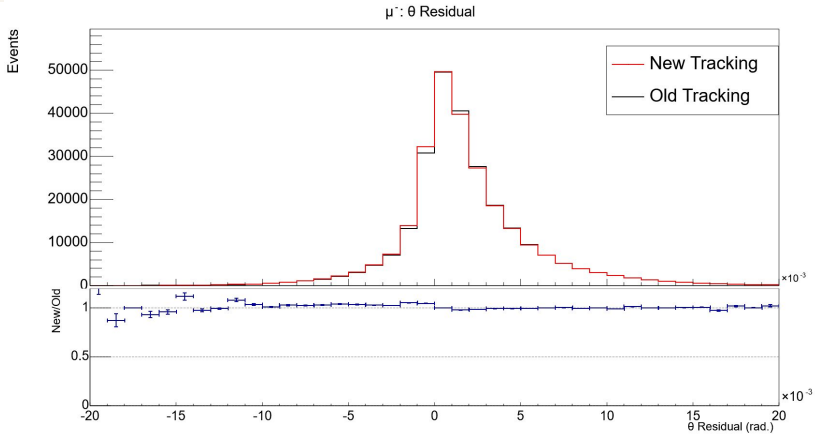


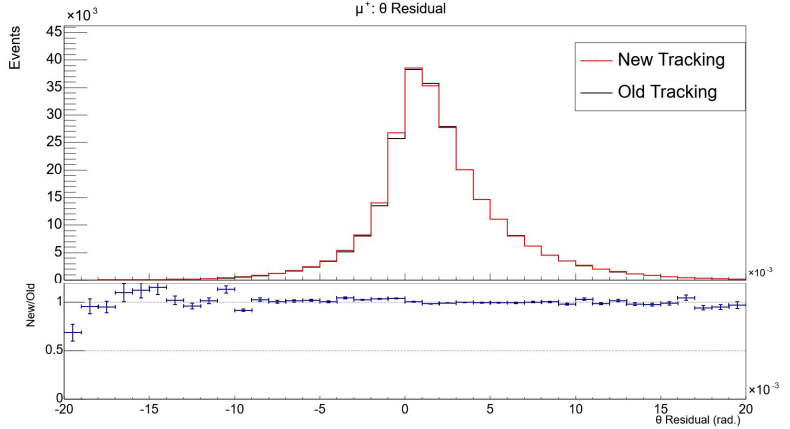
Residuals: Y Momentum



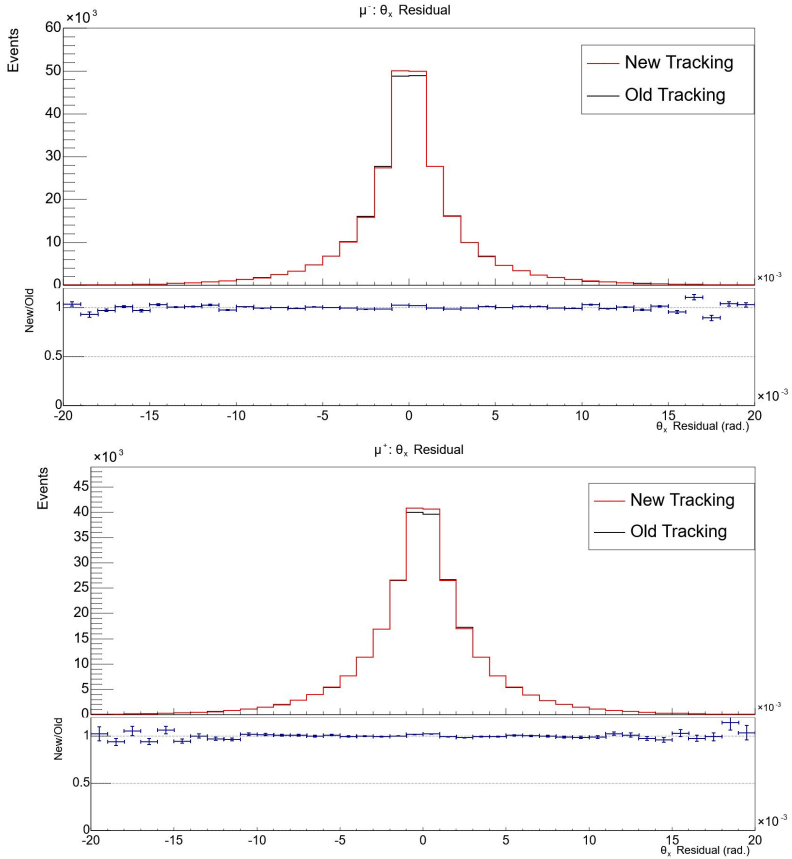


Residuals: Theta

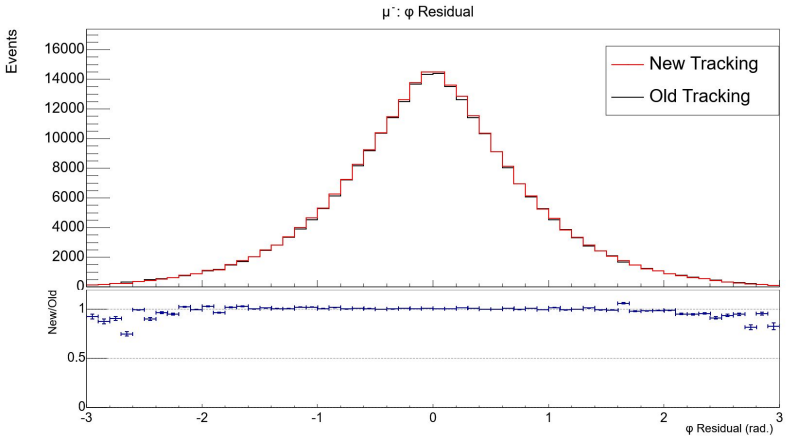


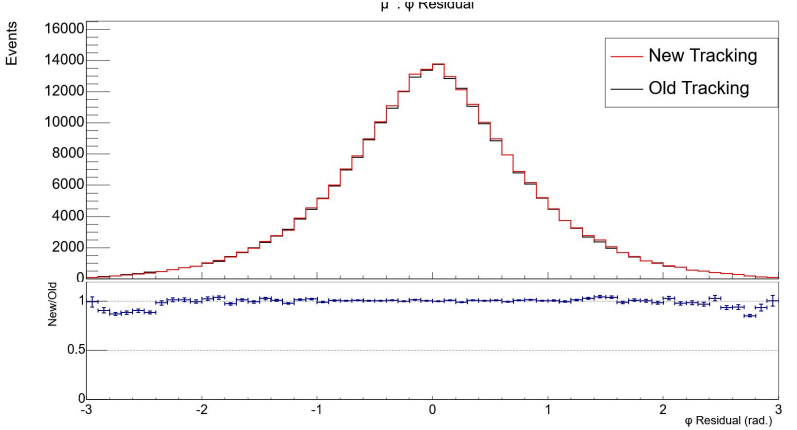


Residuals: Theta X

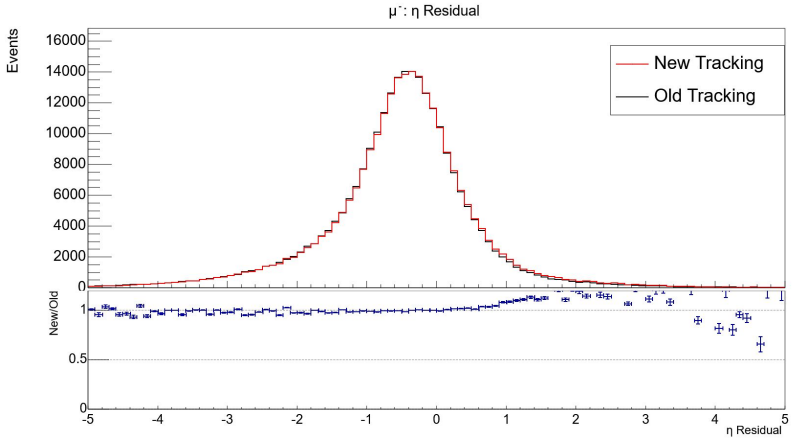


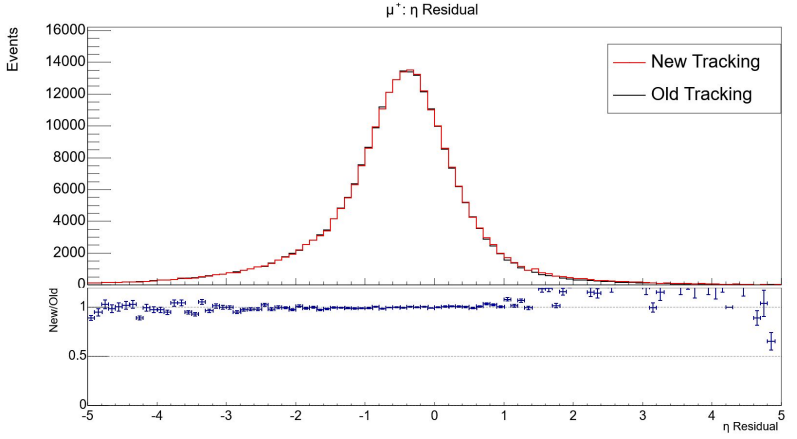
Residuals: Phi





Residuals: Eta

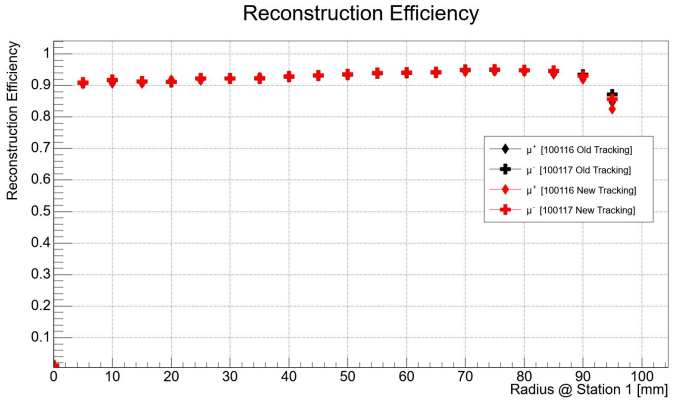


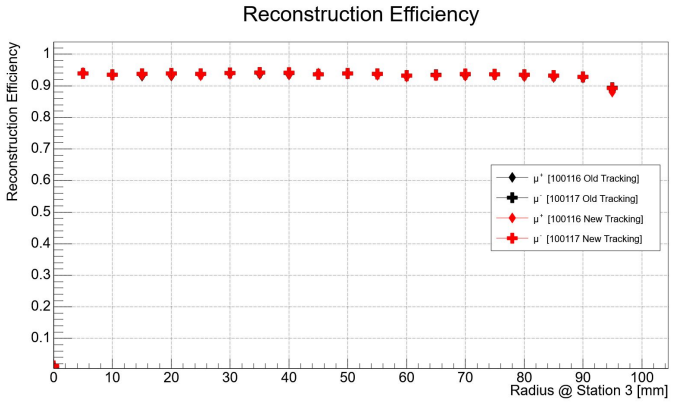


Efficiencies

Reconstruction Efficiency

- Reconstruction efficiency at both station 1 and station 3 of the tracker
  - Front and back of the tracker
  - For each radii - a reconstruction efficiency is calculated

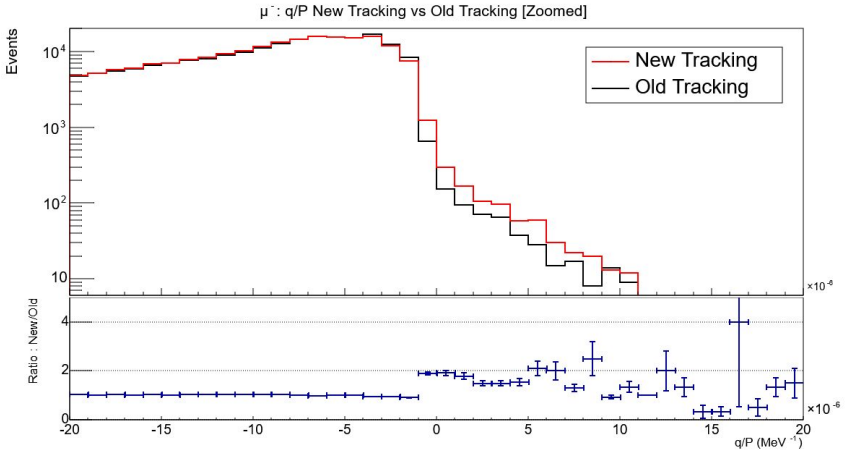


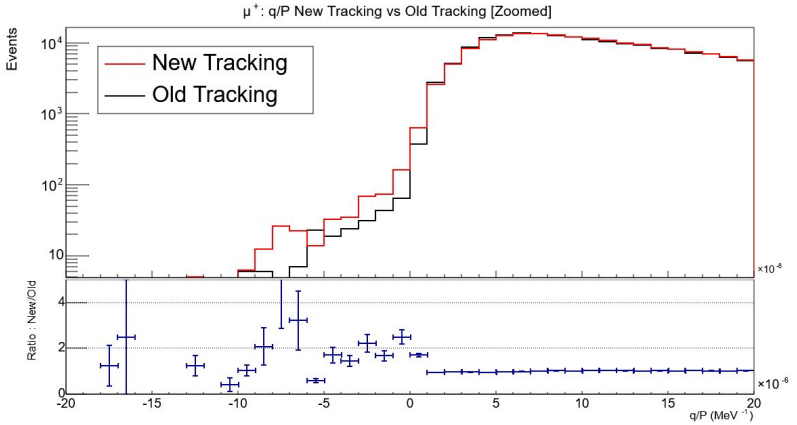


Charge Misidentification

Recap: Charge MisID

- Larger charge misid seen in new tracking
- Looked at q/P to try and find some answers
  - Used track charge/reco momentum
- Ratio [New/old] =1 while charge is correctly identified
  - e.g. q/P is negative for negatively charged particle
- Moving in the region where one would expect events with charge incorrectly identified
  - Increase in events from new tracking vs old tracking



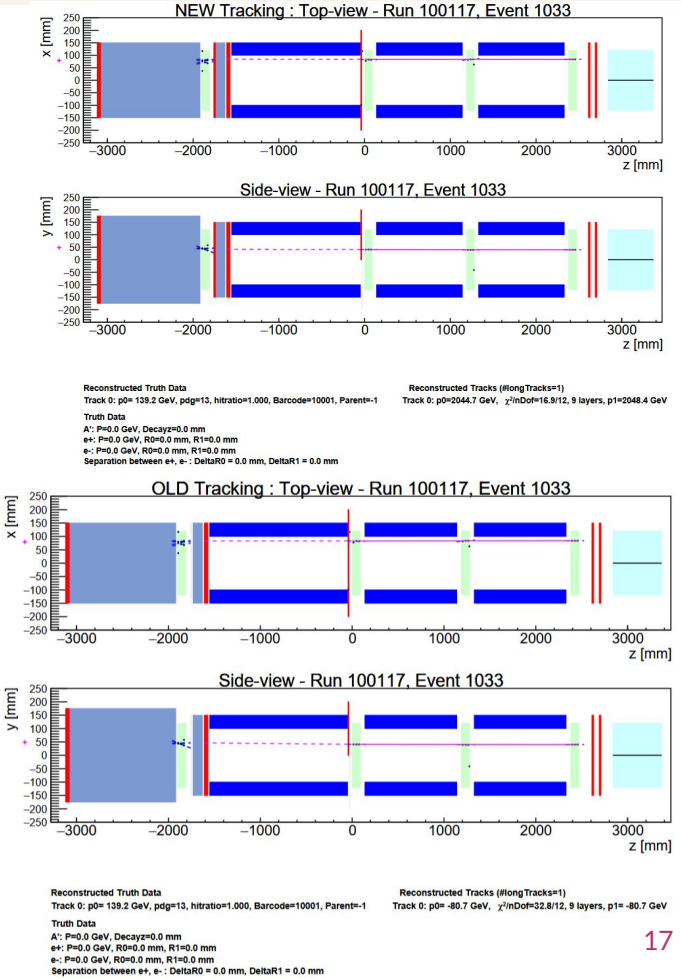


Event Displays (Mu^-)

- Event has charge misidentified using new tracking [top] and charge correctly identified using old tracking [bottom]
- ChargeID percentage =
  - #Events (Fid & 1 GT & Charge ID Correct)/ #Events (Fid & 1 GT)
- ChargeMisID percentage =
  - #Events (Fid & 1 GT & Charge ID Incorrect)/ #Events (Fid & 1 GT)
-

Particle	Tracking	ChargeID (%)	ChargeMisID (%)	Old ChargeID & New ChargeMisID (%)
Mu+	New	99.8	0.197	0.122
	Old	99.9	0.101	
Mu-	New	99.6	0.359	0.223
	Old	99.8	0.217	

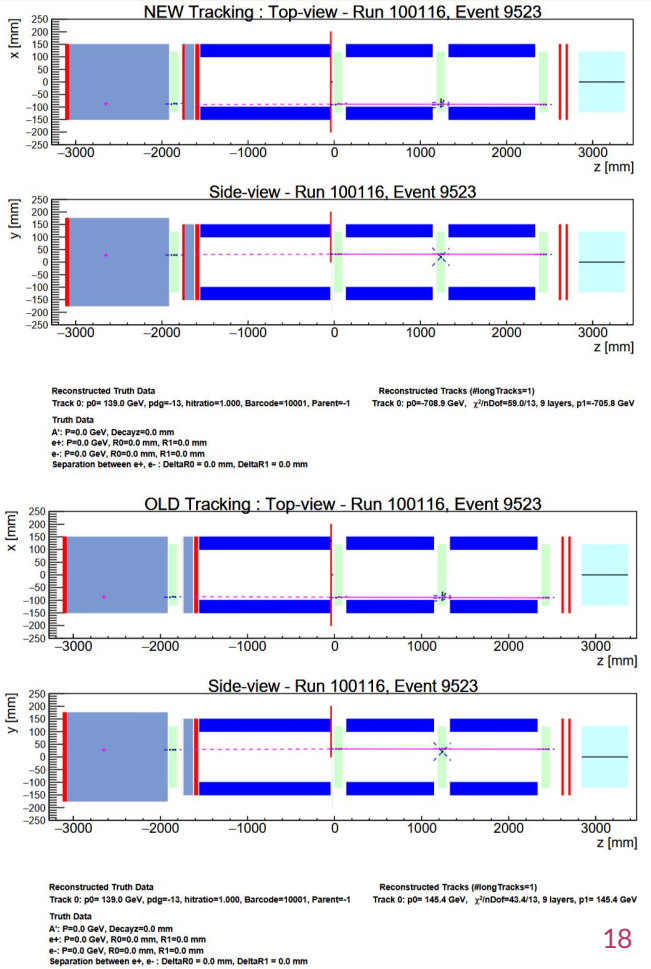
GT : Good Track



Event Displays : (Mu^{+})

- Event has charge misidentified using new tracking [top] and charge correctly identified using old tracking [bottom]
- ChargeID percentage =
  - #Events (Fid & 1 GT & Charge ID Correct)/ #Events (Fid & 1 GT)
- ChargeMisID percentage =
  - #Events (Fid & 1 GT & Charge ID Incorrect)/ #Events (Fid & 1 GT)

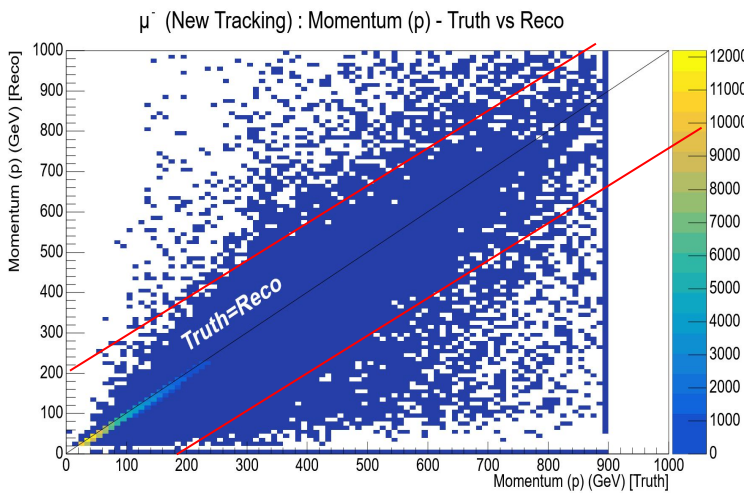
Particle	Tracking	ChargeID (%)	ChargeMisID (%)	Old ChargeID & New ChargeMisID (%)
Mu+	New	99.8	0.197	0.122
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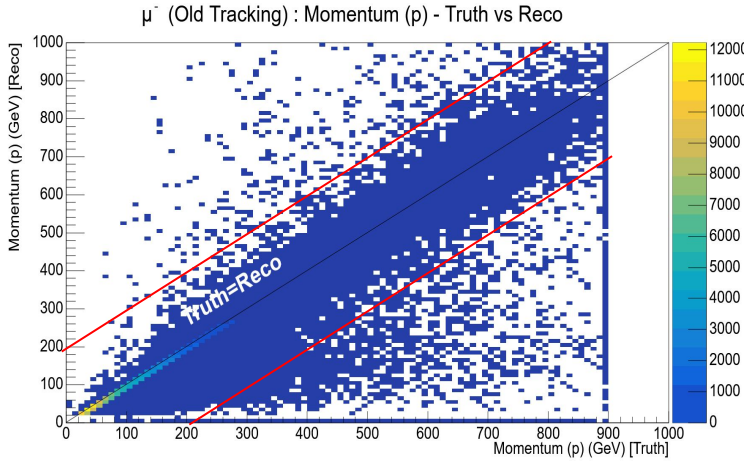


GT : Good Track

Mu^+ : Truth vs Reco (Momentum)

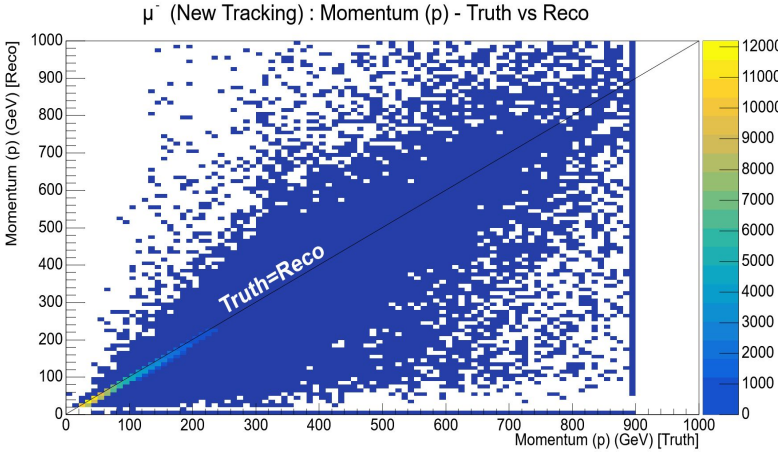
- New tracking appears to have slightly different spread than old tracking
  - I.e. less concentrated around truth=reco line
  - At low momentum
    - Good concentration of truth=reco events
      - Clear correlation seen here

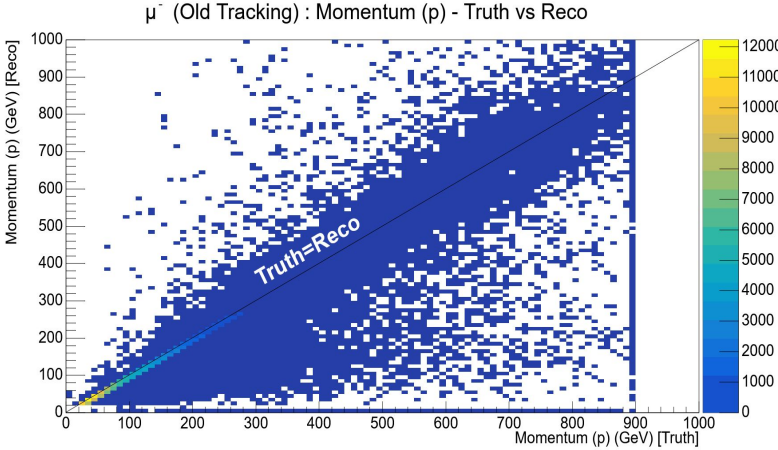




Mu⁻ : Truth vs Reco (Momentum)

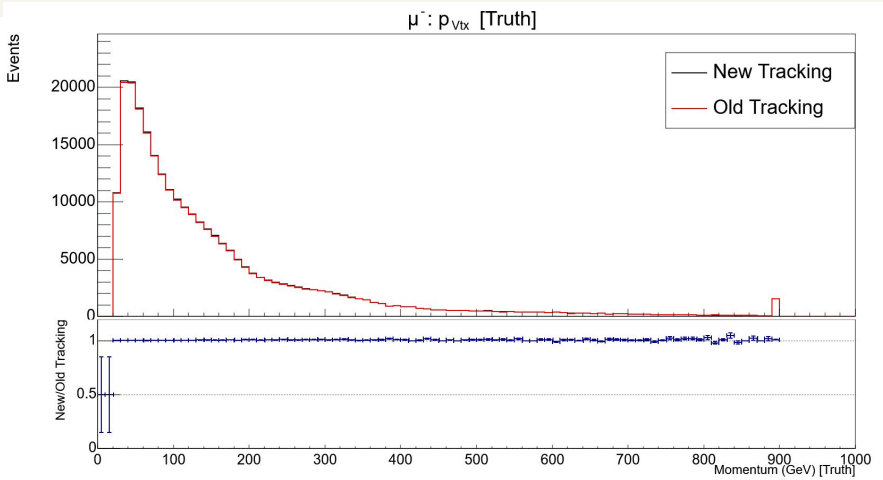
- New tracking appears to have slightly different spread than old tracking
  - I.e. less concentrated around truth=reco line
  - At low momentum
    - Good concentration of truth=reco events
      - Clear correlation seen here

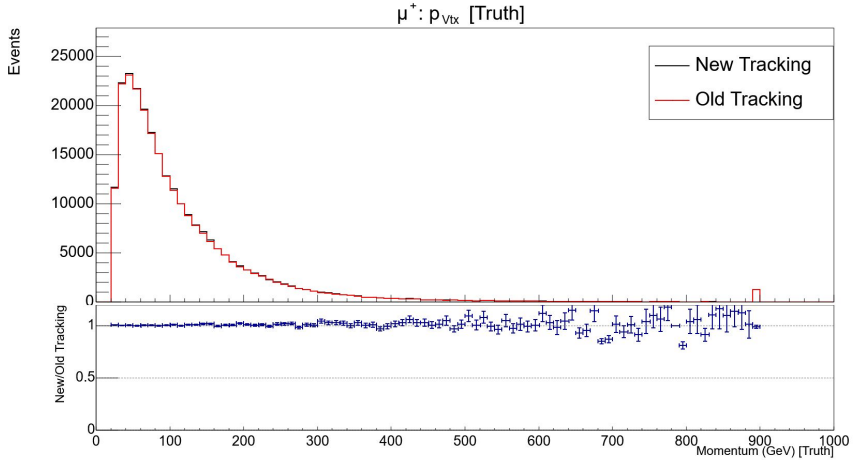




Comparing New vs Old : Truth Momentum

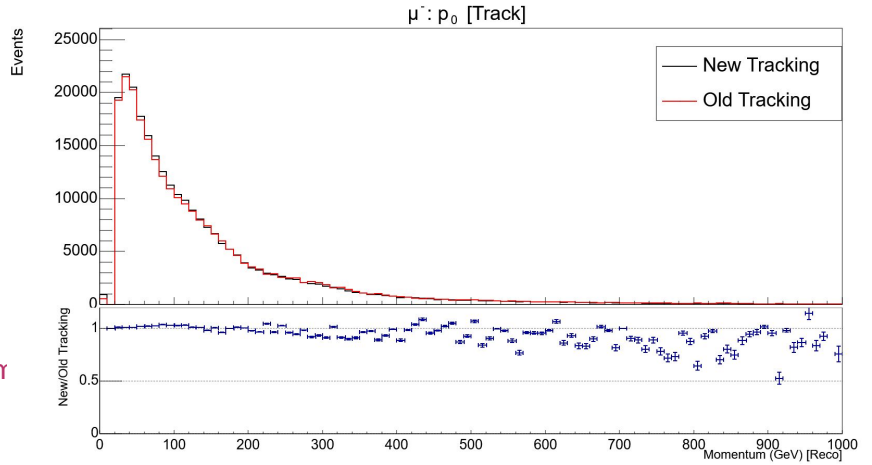
- Agreement between new and old tracking for truth momentum is very good for mu-
  - Seems to be a dip however at very low momentum
- Agreement between new and old tracking for truth momentum good for mu+
  - Gets worse at higher momentum but tends to sit around the ratio ==1

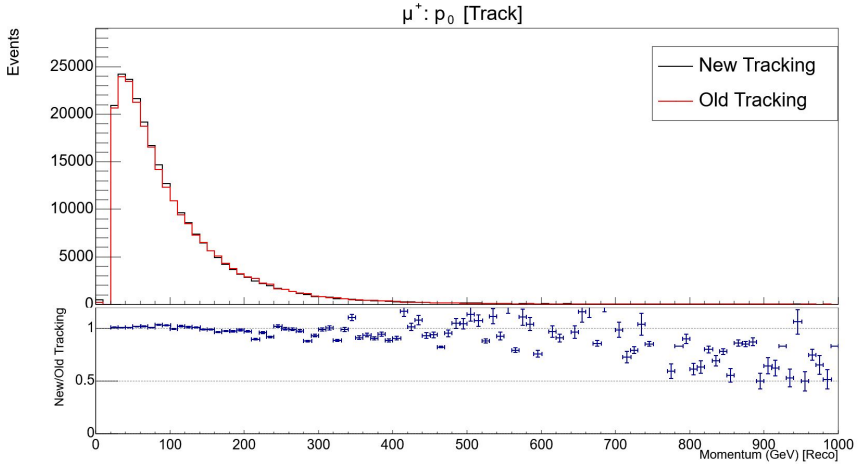




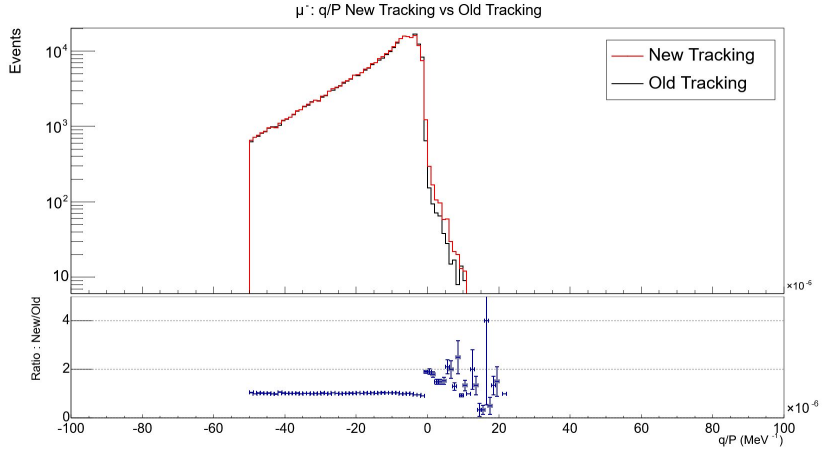
Comparing New vs Old : Track Momentum

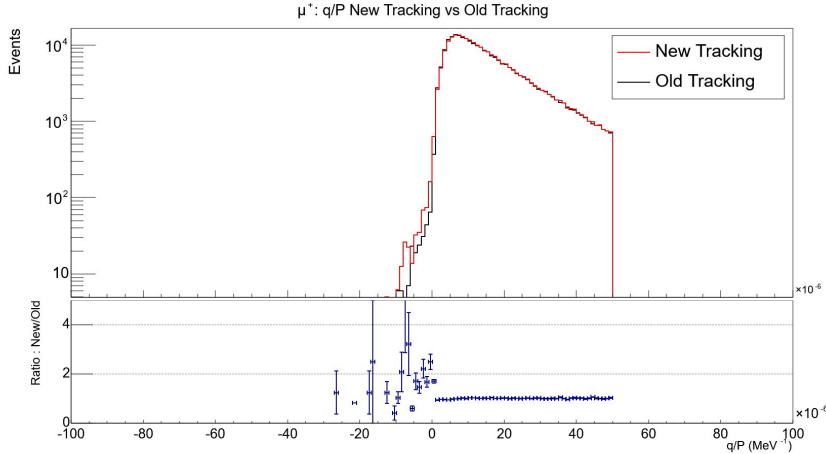
- Reco momentum
  - New and old tracking has good agreement at low momentum < 100
  - Disagreement comes about steadily at momentum of > 200 GeV
    - New tracking appears to have less events than old tracking here



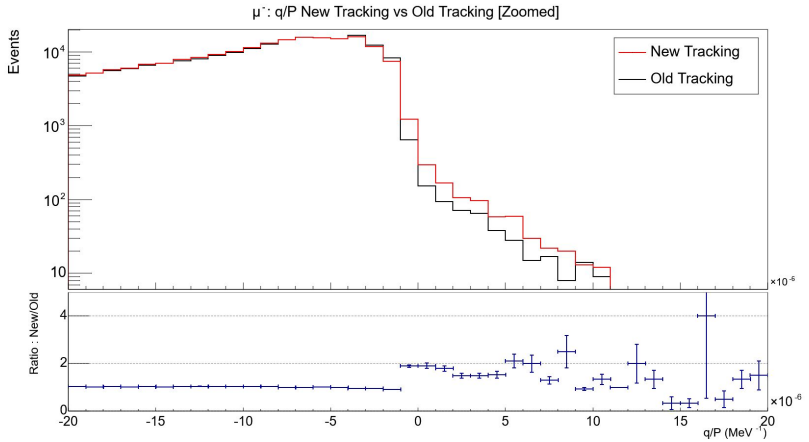


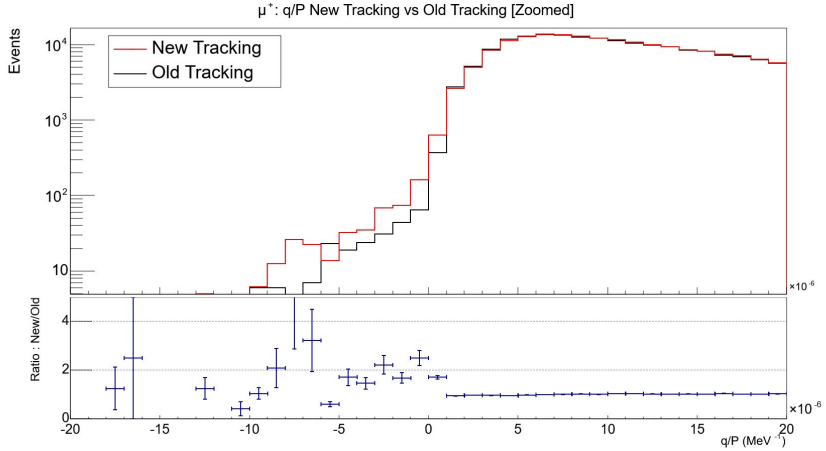
Charge Misidentification



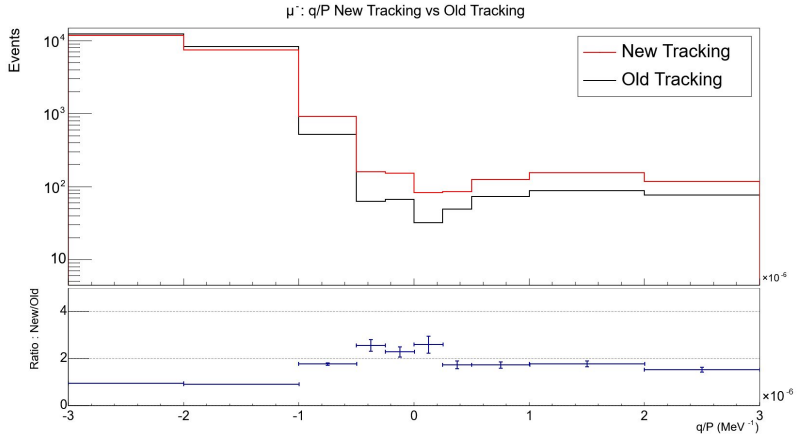


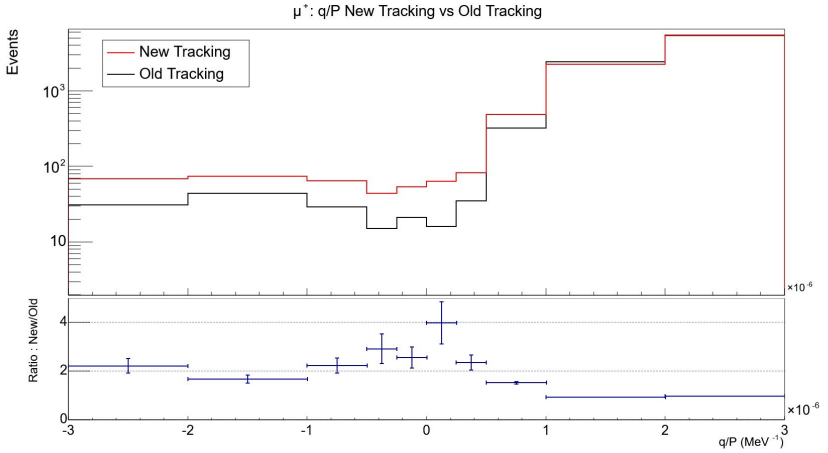
Charge Misidentifcation





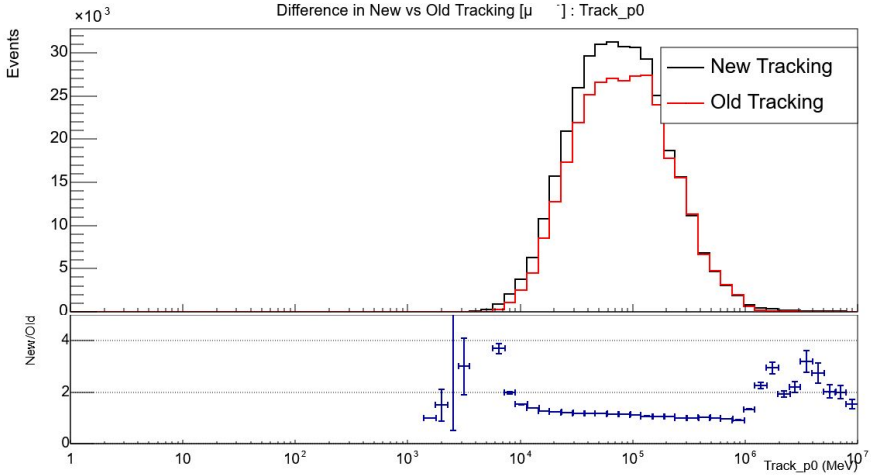
Charge Misidentifcation

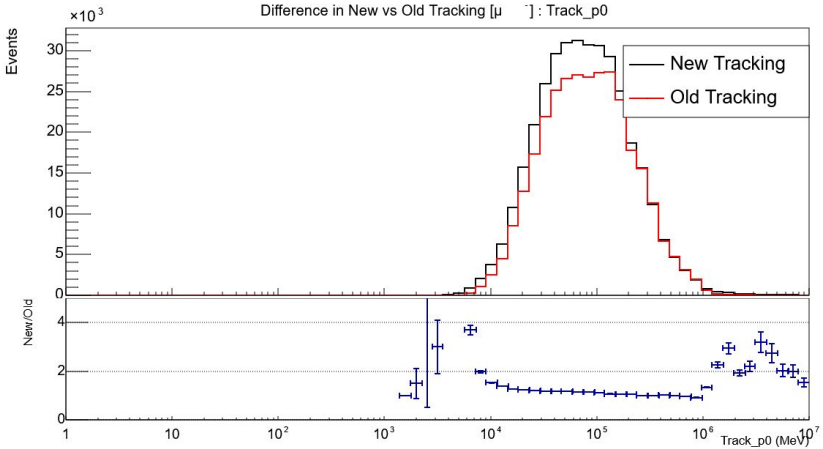




Differences in Old vs New

Track_p0


Difference in New vs Old Tracking [μ] : Track_p0
×10 3
Events
New
Old
New/Old
Track_p0 (MeV)


Difference in New vs Old Tracking [μ] : Track_p0
×10 3
Events
New Tracking
Old Tracking
New/Old
Track_p0 (MeV)